always works, we have a very linear (and unrealistic) workflow.

Listing 11.2 shows these trivial operations; we also define an auxiliary type for shops in listing 11.1: it encapsulates the info associated with a retailer – it will become even more useful later in the chapter, as we develop our final solution. We store the location of the shop in the field `point`, for the sake of consistency with what we have seen in the previous chapters.

Listing 11.2 The addShop and BuyItem methods

```
function addShop(treeRoot, shopName, shopLocation, items=
[])         #1
  shop ← new Shop(shopName, shopLocation, items)
#2
  if treeRoot == null
then                                          #3
    return new Node(shop, null, null,
0)                              #4
  else
    return insert(treeRoot, shop, treeNode.level +
1)                #5

function buyItem(treeRoot, customerLocation,
item)                    #6
  closestShop ← nearestNeighbor(treeRoot,
customerLocation)          #7
  closestShop.order(item,
customerLocation)                           #8
```

#1 The addShop method takes the root of our container tree, the name of the shop and its location (a 2-D point), and